

Project 1:

Scenario-

Your CEO has decided that the company needs a full-time data scientist, and possibly a team of them in the future. She thinks she needs someone who can help drive data science within the entire organization and could potentially lead a team in the future. She understands that data scientist salaries vary widely across the world and is unsure what to pay them. To complicate matters, salaries are going up due to the great recession and the market is highly competitive. Your CEO has asked you to prepare an analysis on data science salaries and provide them with a range to be competitive and get top talent. The position can work offshore, but the CEO would like to know what the difference is for a person working in the United States. Your company is currently a small company but is expanding rapidly.

```
In [11]: import tkinter as tk
from tkinter import filedialog
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

sns.set_theme(style="whitegrid")

def filename(title='Please select a File'):
    root = tk.Tk()
    root.withdraw()
    root.attributes("-topmost", True)
    root.update()
    path = filedialog.askopenfilename(title=title, filetypes=[("CSV files", "*.csv")])
    root.destroy()
    return path

infile = filename()
df = pd.read_csv(infile)
```

What is the market rate for a data scientist?

```
In [12]: print(df.shape)
print(df.columns.tolist())
df.head()

(607, 12)
['Unnamed: 0', 'work_year', 'experience_level', 'employment_type', 'job_title', 'salary', 'salary_currency', 'salary_in_usd', 'employee_residence', 'remote_ratio', 'company_location', 'company_size']
```

Out[12]:

	Unnamed: 0	work_year	experience_level	employment_type	job_title	salary	salary_curr
0	0	2020	MI	FT	Data Scientist	70000	
1	1	2020	SE	FT	Machine Learning Scientist	260000	
2	2	2020	SE	FT	Big Data Engineer	85000	
3	3	2020	MI	FT	Product Data Analyst	20000	
4	4	2020	SE	FT	Machine Learning Engineer	150000	

In [22]:

```
#Making good labels and ensuring that the data is clean and ready
print("Duplicates:", df.duplicated().sum())
df = df.drop_duplicates()
print("Missing values:\n", df.isna().sum())

df["experience"] = df["experience_level"].map(
    {"EN": "Entry", "MI": "Mid", "SE": "Senior", "EX": "Executive"})
df["size"] = df["company_size"].map({"S": "Small", "M": "Medium", "L": "Large"})
df["location"] = np.where(df["employee_residence"] == "US", "U.S.", "Offshore")

df[["work_year", "salary_in_usd"]].describe()
```

```
Duplicates: 0
Missing values:
  Unnamed: 0      0
work_year      0
experience_level  0
employment_type  0
job_title       0
salary          0
salary_currency  0
salary_in_usd   0
employee_residence  0
remote_ratio     0
company_location  0
company_size     0
experience       0
size            0
location        0
dtype: int64
```

Out[22]:

	work_year	salary_in_usd
count	607.000000	607.000000
mean	2021.405272	112297.869852
std	0.692133	70957.259411
min	2020.000000	2859.000000
25%	2021.000000	62726.000000
50%	2022.000000	101570.000000
75%	2022.000000	150000.000000
max	2022.000000	600000.000000

In [23]:

```
#Summary table by experience level order
order = ["Entry", "Mid", "Senior", "Executive"]

df.groupby("experience")["salary_in_usd"].describe().loc[order]
```

Out[23]:

	count	mean	std	min	25%	50%	75%
experience							
Entry	88.0	61643.318182	44395.541126	4000.0	27505.0	56500.0	85425.75
Mid	213.0	87996.056338	63901.057478	2859.0	48000.0	76940.0	112000.00
Senior	280.0	138617.292857	57691.978337	18907.0	100000.0	135500.0	170000.00
Executive	26.0	199392.038462	117071.255697	69741.0	130006.5	171437.5	233750.00

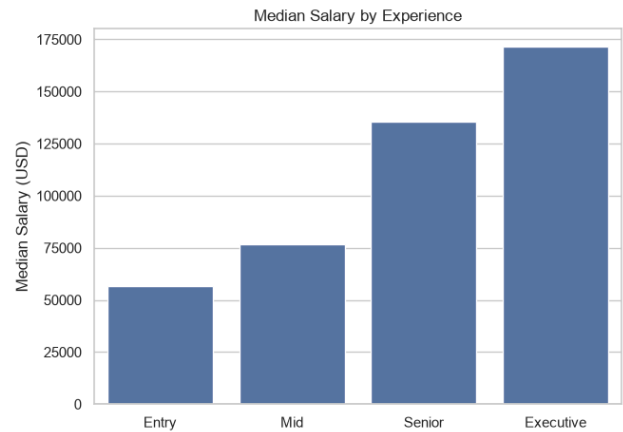
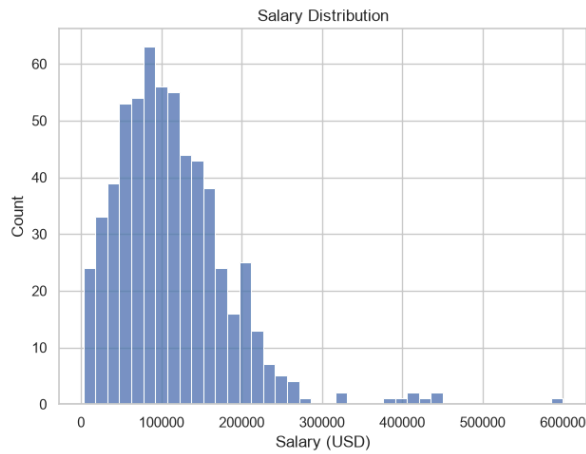
In [15]:

```
fig, ax = plt.subplots(1, 2, figsize=(13, 5))

sns.histplot(df["salary_in_usd"], bins=40, ax=ax[0])
ax[0].set_title("Salary Distribution")
ax[0].set_xlabel("Salary (USD)")

sns.barplot(data=df, x="experience", y="salary_in_usd", order=order,
            estimator=np.median, errorbar=None, ax=ax[1])
ax[1].set_title("Median Salary by Experience")
ax[1].set_xlabel("")
ax[1].set_ylabel("Median Salary (USD)")

plt.tight_layout()
plt.savefig("slide3_baseline.png", dpi=300, bbox_inches="tight")
plt.show()
```



```
In [16]: us = df.loc[df["location"] == "U.S.", "salary_in_usd"]
off = df.loc[df["location"] == "Offshore", "salary_in_usd"]

print(f"U.S. median:      ${us.median():.0f} (n={len(us)})")
print(f"Offshore median:  ${off.median():.0f} (n={len(off)})")
print(f"U.S. premium:     {(us.median() / off.median() - 1):.0%}")

# Salaries are skewed, so Mann-Whitney is safer than a t-test
stat, p = stats.mannwhitneyu(us, off, alternative="greater")
print(f"Mann-Whitney p-value: {p:.2e}")

U.S. median:      $138,475 (n=332)
Offshore median:  $62,649 (n=275)
U.S. premium:     121%
Mann-Whitney p-value: 1.07e-60
```

```
In [17]: # Top countries chart
top = df["employee_residence"].value_counts().head(8).index
sub = df[df["employee_residence"].isin(top)]

plt.figure(figsize=(10, 5))
sns.barplot(data=sub, x="employee_residence", y="salary_in_usd",
            estimator=np.median, errorbar=None,
            order=sub.groupby("employee_residence")["salary_in_usd"].median()
                  .sort_values(ascending=False).index)
plt.title("Median Salary by Country (Top 8 by Number of Records)")
plt.xlabel("Country code")
plt.ylabel("Median Salary (USD)")
plt.savefig("slide4_location.png", dpi=300, bbox_inches="tight")
plt.show()
```



```
In [18]: # Senior and executive roles, small/medium companies
target = df[df["experience"].isin(["Senior", "Executive"]) &
            df["size"].isin(["Small", "Medium"])]

def ranges(group):
    q = group["salary_in_usd"].quantile([0.50, 0.75, 0.90])
    return pd.Series({
        "n": len(group),
        "Competitive (50th-75th)": f"${q[0.50]:,.0f} - ${q[0.75]:,.0f}",
        "Top talent (75th-90th)": f"${q[0.75]:,.0f} - ${q[0.90]:,.0f}",
    })

summary = target.groupby("location").apply(ranges)
summary
```

```
Out[18]:
```

	n	Competitive (50th-75th)	Top talent (75th-90th)
location			
Offshore	48	\$90,968 - \$126,250	\$126,250 - \$164,272
U.S.	175	\$140,000 - \$175,050	\$175,050 - \$210,900

```
In [19]: # Regression: what drives salary? (log salary so coefficients ~ % effect)
import statsmodels.formula.api as smf

model = smf.ols(
    "np.log(salary_in_usd) ~ C(experience, Treatment('Entry')) + C(location) + C(size)",
    data=df
).fit()

print(model.summary().tables[1])
print("R-squared:", round(model.rsquared, 3))
```

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```

			coef	std err	t
P> t	[0.025	0.975]			

Intercept			10.5454	0.080	132.122
0.000	10.389	10.702			
C(experience, Treatment('Entry'))[T.Executive]			1.1177	0.125	8.951
0.000	0.872	1.363			
C(experience, Treatment('Entry'))[T.Mid]			0.3654	0.071	5.113
0.000	0.225	0.506			
C(experience, Treatment('Entry'))[T.Senior]			0.6812	0.072	9.401
0.000	0.539	0.823			
C(location)[T.U.S.]			0.7635	0.051	15.079
0.000	0.664	0.863			
C(size)[T.Medium]			-0.1255	0.051	-2.463
0.014	-0.226	-0.025			
C(size)[T.Small]			-0.1990	0.074	-2.672
0.008	-0.345	-0.053			
remote_ratio			0.0005	0.001	0.970
0.333	-0.001	0.002			

```

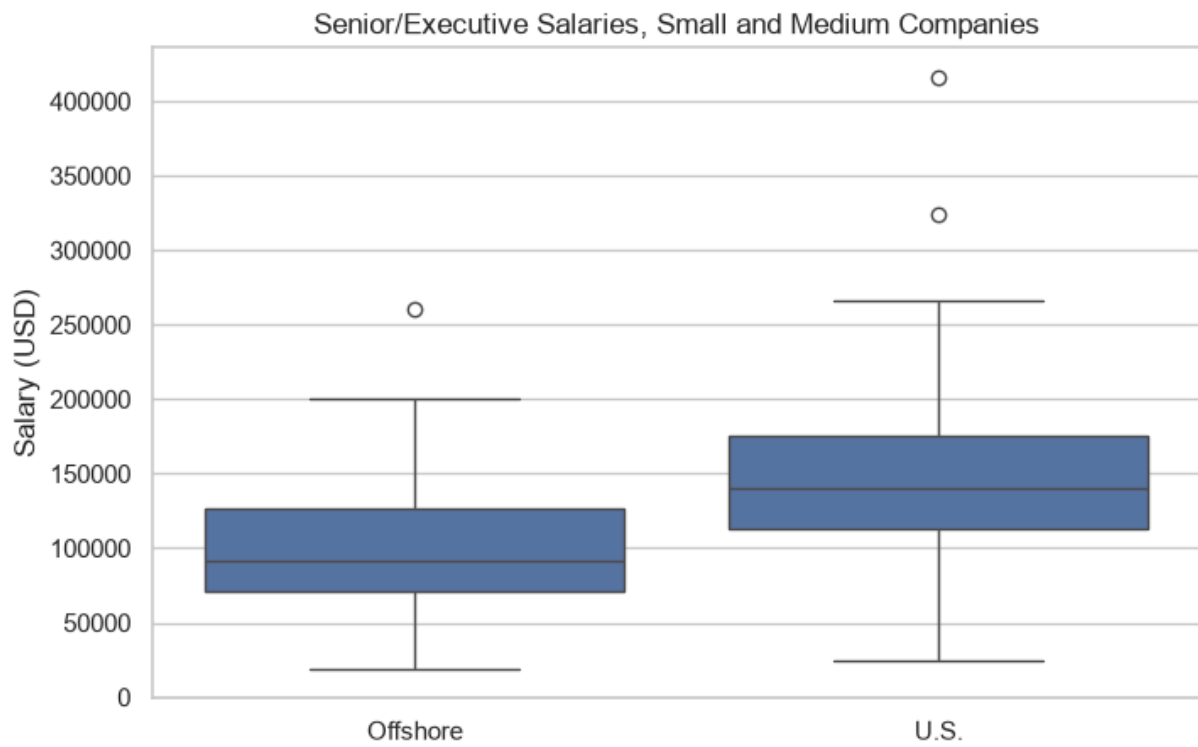
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=====
R-squared: 0.494

```

```

In [20]: # Salary ranges chart
plot_df = target.copy()
plt.figure(figsize=(8, 5))
sns.boxplot(data=plot_df, x="location", y="salary_in_usd")
plt.title("Senior/Executive Salaries, Small and Medium Companies")
plt.xlabel("")
plt.ylabel("Salary (USD)")
plt.savefig("slide5_ranges.png", dpi=300, bbox_inches="tight")
plt.show()

```



```
In [21]: print(f"U.S. premium: {(us.median() / off.median() - 1):.0%}")
         print(summary.to_string())
```

U.S. premium: 121%

	n	Competitive (50th-75th)	Top talent (75th-90th)
location			
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In []: